Anisotropic Impedance Control for Valve Manipulation under Unmodeled Support Motion

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***Abstract*—Contact manipulation requires controlled motion along task directions while forces and motion in the remaining directions stay compatible with contact geometry. Mechanical clearance, support compliance, wear, deformation, and external loading can introduce motion of the contacted mechanism that is absent from the controller model. This paper studies how task-aligned anisotropic impedance affects valve tracking when the valve support moves through an unmeasured passive coordinate. A simulated Unitree Z1 arm mounted on a B2 quadruped is controlled either by a diagonal Cartesian impedance or by an anisotropic formulation that separates valve progress from the remaining task directions. Lower and higher valve-direction gain settings are evaluated, and an additional anisotropic configuration uses the lower gain level also in the remaining task directions. The disturbed condition applies the same bipolar support torque, with peak magnitude 3.5 N m, while the support stiffness and damping remain unchanged. With the lower setting, disturbed tracking RMSE is 0.444° for CART-L, 0.159° for ANISO-LL, and 0.145° for ANISO-L. With higher gains, CART-H and ANISO-H reach 0.135° and 0.138°, respectively. The results show that the lower tracking error of the task-aligned formulation persists without higher gains in the remaining task directions, while its response depends on the selected gain distribution.**

***Index Terms*—anisotropic impedance, operational-space control, constrained manipulation, environmental compliance, valve manipulation**

I. Introduction

Physical interaction tasks are governed simultaneously by motion requirements and by constraints imposed by contact geometry. These constraints are not always rigid. Mechanical clearance, wear, passive compliance, structural deformation, and external loading can allow the contacted mechanism to move in directions that are absent from its nominal model. The robot must then continue the intended motion while accommodating changes in the interaction geometry and the associated forces.

Impedance control establishes a dynamic relation between motion and interaction wrench [1], and operational-space formulations express this relation directly in task coordinates [2], [3]. Task-frame methods provide a geometric basis for separating directions according to their role in an interaction [4]–[6]. Anisotropic impedance extends this idea by assigning different stiffness and damping along different task directions, as used in task-dependent guidance and constrained manipulation [7]–[9].

When the contacted mechanism moves through a passive coordinate that is absent from the controller model, the relative pose between robot and environment changes during interaction. Under these conditions, the distribution of impedance among task directions affects how this motion propagates into the commanded task. Lower impedance along the desired motion can increase compliance, whereas stronger impedance in the remaining task directions can help preserve alignment.

Valve manipulation provides a natural separation of these roles. After grasping, rotation about the valve axis defines progress toward the task objective, while the remaining Cartesian directions maintain grasp compatibility. Previous work has studied robotic valve turning with different platforms and compliance strategies [10], [11]. The valve scenario is used here as a representative constrained-interaction example in which anisotropic impedance can be evaluated under unmodeled motion of the contacted mechanism. In the present setup, the valve housing is mounted through a passive support coordinate that belongs to the environment rather than to the robot. An external torque moves this support during the turn without exposing its state or the applied torque to the arm controller.

This paper compares a diagonal Cartesian impedance controller (CART) with a task-aligned anisotropic formulation (ANISO). The main comparison evaluates lower and higher

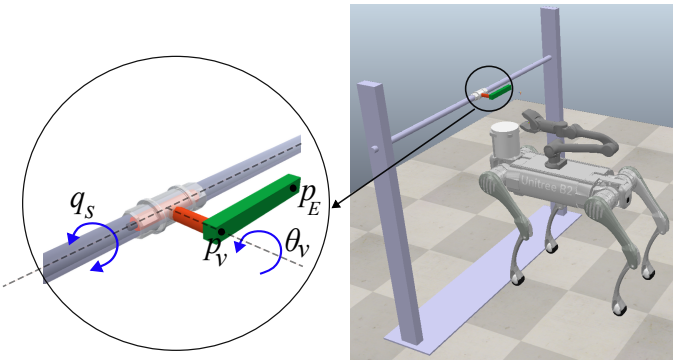


Fig. 1. Simulated B2–Z1 system and valve support.

gains along the valve-progress direction, denoted by the suffixes L and H. At each setting, the corresponding CART and ANISO configurations have matched nominal stiffness and damping along this direction at the initial valve configuration. ANISO-L and ANISO-H use the higher gain level in the remaining task directions. An additional configuration, ANISO-LL, uses the lower gain level in both the valve-progress and remaining task directions to examine whether the behavior observed at the lower setting persists without higher complementary gains. All configurations are evaluated under nominal and disturbed support conditions to examine how environmental motion appears in valve tracking and in the remaining task directions.

II. System and Experimental Setup

A. Robot and manipulation task

The system was implemented in CoppeliaSim and combines a Unitree B2 quadruped, a six-degree-of-freedom Unitree Z1 manipulator, a two-jaw gripper, a wrist force/torque sensor, and a revolute valve. The force/torque sensor was used for supervision and evaluation but did not enter the impedance feedback law. The B2 leg controller maintains stance while the impedance controller commands the six Z1 joints. The arm torque limits are

$$\boldsymbol{\tau}_{\max} = [30,\ 60,\ 30,\ 30,\ 30,\ 30]^{\mathsf{T}}\ \mathrm{N\,m}. \tag{1}$$

The valve joint was modeled as a passive revolute joint with a 0–90° range and no additional joint actuation or damping. The valve handle was a 0.5 kg dynamic body with a local inertia tensor about its center of mass of $\mathrm{diag}(0.01044,\ 0.01417,\ 0.00377)\ \mathrm{kg\,m^2}$. Grasping was simulated through Bullet contact between the handle and the two gripper jaws, with friction coefficients of 0.5 for the handle and moving jaw and 0.71 for the fixed jaw. Restitution and additional linear and angular damping were set to zero. The grasp was maintained by the dynamically controlled gripper joint, with a maximum motor torque of 3 N m.

All control quantities are expressed in the arm mounting frame $\{B\}$. The arm joint position and velocity are used to compute the model-based terms and Jacobian.

The measured tool center point (TCP) pose and velocity are used to compute the task-space errors $\bar{\boldsymbol{e}}_c$ and $\bar{\tilde{\boldsymbol{\nu}}}_c$. The measured valve angle θ_v defines the valve-tracking error, while the valve pivot $\boldsymbol{p}_v^0$ and axis $\boldsymbol{a}_v^0$, recorded at the onset of rotation, define the valve-progress direction. These reference quantities remain fixed throughout the turn and are not updated from the support motion.

B. Valve-progress and remaining task directions

Let $\boldsymbol{p}_v^0$ and $\boldsymbol{a}_v^0$ denote the valve pivot and unit rotation axis at the onset of valve motion, and let $\boldsymbol{p}_E$ be the TCP position. With $\boldsymbol{r} = \boldsymbol{p}_E - \boldsymbol{p}_v^0$, the TCP twist associated with unit valve rotation is

$$\boldsymbol{\xi}_v = \begin{bmatrix} \boldsymbol{a}_v^0 \times \boldsymbol{r} \\ \boldsymbol{a}_v^0 \end{bmatrix}. \tag{2}$$

Translation and rotation are placed on a common task-space scale using $L_x = 0.300$ m. This value was selected heuristically as a fixed scaling parameter for the simulated task and was not optimized for any controller configuration. The characteristic length sets the relative weighting between linear and angular components; an angular error $\delta\phi$ contributes $L_x\delta\phi$ equivalent metres in the scaled coordinates. Because it also enters the scaled valve twist, its value affects the relative translational–rotational weighting of the valve-progress direction and its orthogonal complement. Sensitivity to this parameter was not evaluated in the present study. Thus,

$$\boldsymbol{S}_x = \mathrm{diag}(\boldsymbol{I}_3, L_x\boldsymbol{I}_3), \qquad \bar{\boldsymbol{e}} = \boldsymbol{S}_x\boldsymbol{e}, \quad \bar{\boldsymbol{J}} = \boldsymbol{S}_x\boldsymbol{J}. \tag{3}$$

The scaled valve twist $\bar{\boldsymbol{\xi}}_v = \boldsymbol{S}_x\boldsymbol{\xi}_v$ defines the valve-progress direction

$$\boldsymbol{u}_v = \frac{\bar{\boldsymbol{\xi}}_v}{\|\bar{\boldsymbol{\xi}}_v\|}, \qquad \boldsymbol{P}_\perp = \boldsymbol{I}_6 - \boldsymbol{u}_v\boldsymbol{u}_v^{\mathsf{T}}. \tag{4}$$

The unit vector $\boldsymbol{u}_v$ spans the valve-progress direction, while $\boldsymbol{P}_\perp$ spans its five-dimensional orthogonal complement, which is regulated to preserve the grasp geometry. Motion in this subspace does not advance the ideal valve rotation; instead, it changes the TCP pose relative to the grasp and captures alignment changes induced by support motion.

The valve reference commands an 88° rotation over 24 s with smooth zero-velocity boundary conditions. A compatible TCP reference for the remaining task directions is reconstructed from the measured valve angle and the reference valve geometry, while valve progress is regulated separately through e_θ. The reference is based on the valve geometry measured at the beginning of the turn. The valve-tracking error is

$$e_\theta = \theta_d - \theta_v, \qquad \dot{e}_\theta = \dot{\theta}_d - \dot{\theta}_v. \tag{5}$$

For compatible task-space pose and velocity errors $\bar{\boldsymbol{e}}_c$ and $\bar{\tilde{\boldsymbol{\nu}}}_c$, the components outside the valve-progress direction are $\bar{\boldsymbol{e}}_\perp = \boldsymbol{P}_\perp\bar{\boldsymbol{e}}_c$ and $\bar{\tilde{\boldsymbol{\nu}}}_\perp = \boldsymbol{P}_\perp\bar{\tilde{\boldsymbol{\nu}}}_c$.

C. Passive support and external disturbance

The valve housing is connected to the fixed structure through an unactuated revolute support joint. During the analyzed interval its restoring torque is

$$\tau_s = -K_s q_s - C_s \dot{q}_s, \tag{6}$$

with $K_s = 67.4486$ N m/rad and $C_s = 4.9957$ N m s/rad. The support axis is approximately orthogonal to the valve axis and its limits are $\pm 10^\circ$.

In the nominal condition no external torque is applied to the support. In the disturbed condition, K_s and C_s remain unchanged while a prescribed torque acts from 7 to 19 s after valve-turn onset. It evolves smoothly from zero to -3.5 N m, reverses to $+3.5$ N m, and returns to zero. The same torque profile is used in all disturbed simulations. This produces repeatable motion of the environmental support while leaving the robot controller unchanged. Because the support joint remains passive in both conditions, interaction forces can produce support deflection even without the externally applied disturbance.

The support coordinate is part of the environment and is not included in the arm controller. The simulations use Bullet dynamics at 800 Hz, with the model terms and arm torque commands updated at 20 Hz and 200 Hz, respectively. The simulation scene is available in the project repository.

III. Impedance Control

A. Directional impedance in task space

A Cartesian impedance law maps task-space pose and velocity errors into a corrective wrench through stiffness and damping. When the same mechanical response is not appropriate in every direction, these gains can instead be organized according to the task coordinates [7], [8]. For the valve task, the projector in (4) gives

$$\bar{\boldsymbol{e}}_c = \boldsymbol{u}_v \boldsymbol{u}_v^{\mathsf{T}} \bar{\boldsymbol{e}}_c + \boldsymbol{P}_\perp \bar{\boldsymbol{e}}_c, \tag{7}$$

with an analogous expression for velocity. The first term advances the valve, whereas the second contains errors in the remaining task directions that maintain the grasp geometry. The ANISO controller assigns these two parts different impedance responses. The CART controller instead applies one diagonal Cartesian gain map to the combined error before the wrench is projected onto the valve-progress direction and its orthogonal subspace.

B. Common operational-space dynamics

The arm dynamics are written as

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} + \boldsymbol{J}^{\mathsf{T}} \boldsymbol{w}_{\mathrm{ext}} + \boldsymbol{d}_B, \tag{8}$$

where $\boldsymbol{M}$, $\boldsymbol{c}$, and $\boldsymbol{g}$ are the arm inertia, Coriolis/centrifugal, and gravity terms, respectively, $\boldsymbol{w}_{\mathrm{ext}}$ is the external task-space wrench, and $\boldsymbol{d}_B$ collects generalized terms not represented by the fixed-base arm model. The regularized operational-space inertia and common model-based term are

$$\begin{aligned} \bar{\boldsymbol{\Lambda}}_\lambda &= (\bar{\boldsymbol{J}} \boldsymbol{M}^{-1} \bar{\boldsymbol{J}}^{\mathsf{T}} + \lambda^2 \boldsymbol{I}_6)^{-1}, \\ \bar{\boldsymbol{w}}_{\mathrm{ff}} &= \bar{\boldsymbol{\Lambda}}_\lambda (\ddot{\bar{\boldsymbol{x}}}_d - \dot{\bar{\boldsymbol{J}}} \dot{\boldsymbol{q}}), \end{aligned} \tag{9}$$

where $\ddot{\bar{\boldsymbol{x}}}_d$ is the scaled desired task-space acceleration and $\lambda = 0.020$. This model-based term is common to both controllers and their difference lies in the organization of the impedance feedback.

C. Diagonal Cartesian impedance

For a valve-direction gain setting $j \in \{L, H\}$, the CART controller forms the combined task-space error

$$\bar{\boldsymbol{e}}_I = \bar{\boldsymbol{e}}_\perp + \bar{\boldsymbol{\xi}}_v e_\theta, \qquad \tilde{\bar{\boldsymbol{\nu}}}_I = \tilde{\bar{\boldsymbol{\nu}}}_\perp + \bar{\boldsymbol{\xi}}_v \dot{e}_\theta, \tag{10}$$

and applies the diagonal Cartesian impedance

$$\bar{\boldsymbol{w}}_{C,j} = \bar{\boldsymbol{K}}_j \bar{\boldsymbol{e}}_I + \bar{\boldsymbol{D}}_j \tilde{\bar{\boldsymbol{\nu}}}_I. \tag{11}$$

The CART feedback wrench is decomposed into the valve-progress and orthogonal components as

$$\bar{\boldsymbol{w}}_{\perp,C} = \boldsymbol{P}_\perp \bar{\boldsymbol{w}}_{C,j}, \qquad \bar{\boldsymbol{w}}_{v,C} = \boldsymbol{u}_v \boldsymbol{u}_v^{\mathsf{T}} \bar{\boldsymbol{w}}_{C,j}. \tag{12}$$

The corresponding valve-direction feedback torque is

$$\tau_{v,C} = \bar{\boldsymbol{\xi}}_v^{\mathsf{T}} \bar{\boldsymbol{w}}_{C,j}. \tag{13}$$

Expanding this torque separates the contribution generated by the valve-tracking error from that produced by errors in the orthogonal task subspace,

$$\tau_{v,C} = \tau_{v,\mathrm{pure}} + \tau_{v,\mathrm{leak}}, \tag{14}$$

where

$$\begin{aligned} \tau_{v,\mathrm{pure}} &= \bar{\boldsymbol{\xi}}_v^{\mathsf{T}} \left(\bar{\boldsymbol{K}}_j \bar{\boldsymbol{\xi}}_v e_\theta + \bar{\boldsymbol{D}}_j \bar{\boldsymbol{\xi}}_v \dot{e}_\theta \right), \\ \tau_{v,\mathrm{leak}} &= \bar{\boldsymbol{\xi}}_v^{\mathsf{T}} \left(\bar{\boldsymbol{K}}_j \bar{\boldsymbol{e}}_\perp + \bar{\boldsymbol{D}}_j \tilde{\bar{\boldsymbol{\nu}}}_\perp \right). \end{aligned} \tag{15}$$

Although $\bar{\boldsymbol{e}}_\perp$ and $\tilde{\bar{\boldsymbol{\nu}}}_\perp$ lie outside the valve-progress direction, the diagonal Cartesian gain matrices can map these errors into a wrench with a nonzero valve-progress component.

D. Task-aligned anisotropic impedance

The ANISO controller assigns the valve-progress impedance directly,

$$\tau_{v,A,j} = K_{\theta,j} e_\theta + D_{\theta,j} \dot{e}_\theta, \tag{16}$$

where the index j defines the valve-direction gain level.

The remaining task directions use either the lower or higher Cartesian gain matrices,

$$\bar{\boldsymbol{w}}_{\perp,A} = \boldsymbol{P}_\perp \left(\bar{\boldsymbol{K}}_k \bar{\boldsymbol{e}}_\perp + \bar{\boldsymbol{D}}_k \tilde{\bar{\boldsymbol{\nu}}}_\perp \right), \tag{17}$$

where the index k defines the gain level in these directions.

The valve-direction wrench is

$$\bar{\boldsymbol{w}}_{v,A} = \frac{\boldsymbol{u}_v}{\left\| \bar{\boldsymbol{\xi}}_v \right\|} \tau_{v,A,j}. \tag{18}$$

The projection in (17) prevents this branch from producing a component along the valve-progress direction. The task-aligned formulation regulates valve progress and the orthogonal task subspace through separate feedback branches rather than through a single Cartesian gain map.

ANISO-L uses lower valve-direction gains and higher gains in the remaining directions. ANISO-H uses the higher gain level in both parts. The additional ANISO-LL configuration uses the lower gain level in both parts.

TABLE I
CONTROLLER CONFIGURATIONS AND GAIN SETTINGS.

Controller	Valve gains	Remaining gains	K_θ (N m/rad)	D_θ (N m s/rad)
CART-L	Lower	Lower	1.50	0.859868
ANISO-LL	Lower	Lower	1.50	0.859868
ANISO-L	Lower	Higher	1.50	0.859868
CART-H	Higher	Higher	3.00	1.719736
ANISO-H	Higher	Higher	3.00	1.719736

E. Gain selection and torque command

The lower valve-direction gains correspond to the baseline impedance used for the valve-turning task. The higher setting doubles both values to evaluate the influence of gain level,

$$\begin{aligned} K_{\theta,L} &= 1.50, \quad D_{\theta,L} = 0.859868, \\ K_{\theta,H} &= 3.00, \quad D_{\theta,H} = 1.719736. \end{aligned} \tag{19}$$

The stiffness values are given in N m/rad and the damping values in N m s/rad. The Cartesian matrix entries were not selected independently.

The relative weighting between translational and rotational terms was retained from the baseline Cartesian tuning, while one common scale factor was applied to stiffness and another to damping. These factors were chosen so that the resulting Cartesian controller had the prescribed valve-direction stiffness and damping at the initial valve configuration. This gives

$$\begin{aligned} \bar{\boldsymbol{K}}_L &= \mathrm{diag}\left(299.877\boldsymbol{I}_3, \frac{1.4994}{L_x^2}\boldsymbol{I}_3\right), \\ \bar{\boldsymbol{D}}_L &= \mathrm{diag}\left(121.636\boldsymbol{I}_3, \frac{0.85962}{L_x^2}\boldsymbol{I}_3\right). \end{aligned} \tag{20}$$

The normalization satisfies

$$\bar{\boldsymbol{\xi}}_{v,0}^{\mathsf{T}}\bar{\boldsymbol{K}}_L\bar{\boldsymbol{\xi}}_{v,0} \approx K_{\theta,L}, \qquad \bar{\boldsymbol{\xi}}_{v,0}^{\mathsf{T}}\bar{\boldsymbol{D}}_L\bar{\boldsymbol{\xi}}_{v,0} \approx D_{\theta,L}. \tag{21}$$

The higher-gain Cartesian controller then uses twice these matrix values,

$$\bar{\boldsymbol{K}}_H = 2\bar{\boldsymbol{K}}_L, \qquad \bar{\boldsymbol{D}}_H = 2\bar{\boldsymbol{D}}_L. \tag{22}$$

At the initial valve configuration, CART-L is matched to the lower valve-direction stiffness and damping, while CART-H is matched to the higher setting. The ANISO controllers apply these valve-direction gains directly.

CART-L and ANISO-LL use the lower gain level along valve progress and in the remaining task directions, allowing the two feedback formulations to be compared at the same gain level. ANISO-LL and ANISO-L use the same task-aligned formulation and lower valve-direction gains, while the remaining task directions use lower and higher gains, respectively. At the higher setting, CART-H and ANISO-H use the higher gain level in both parts. The arm torque command is

$$\boldsymbol{\tau}_{\mathrm{cmd}} = \boldsymbol{g} + \mathrm{sat}_{\boldsymbol{\tau}_d}\left(\boldsymbol{c} + \bar{\boldsymbol{J}}^{\mathsf{T}}(\bar{\boldsymbol{w}}_{\mathrm{ff}} + \bar{\boldsymbol{w}}_{\perp})\right) + s_v\bar{\boldsymbol{J}}^{\mathsf{T}}\bar{\boldsymbol{w}}_v, \tag{23}$$

where $(\bar{\boldsymbol{w}}_{\perp}, \bar{\boldsymbol{w}}_v)$ are supplied by CART or ANISO, $\mathrm{sat}_{\boldsymbol{\tau}_d}$ denotes the per-joint saturation of the non-gravity dynamic term, and $s_v \in [0,1]$ scales the valve-direction contribution according to the available actuator torque margin.

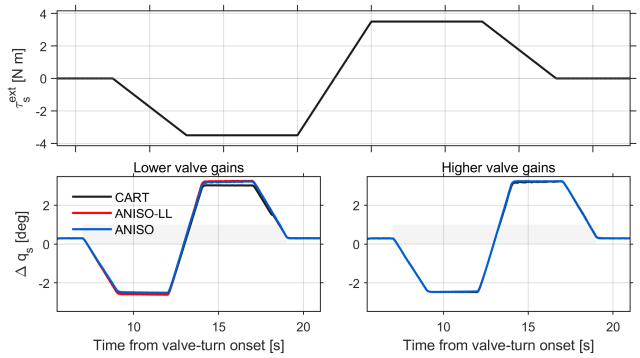


Fig. 2. External support torque and support deflection in the disturbed cases.

IV. EXPERIMENTAL EVALUATION

Five controller configurations are evaluated under nominal and disturbed support conditions. CART-L, ANISO-L, CART-H, and ANISO-H form the main comparison, while ANISO-LL provides the additional low-gain comparison. All simulations start from the same initial state, with identical grasp, valve reference, support parameters, actuator limits, and numerical settings. No external torque is applied to the support in the nominal condition, whereas all disturbed cases use the same prescribed torque profile. One deterministic simulation is performed for each controller-condition combination, resulting in ten simulations.

The primary analysis interval extends from 5.5 to 21 s after valve-turn onset. This interval begins after the initial support transition and ends before the final transition, while containing the complete 7–19 s external disturbance. It therefore includes 1.5 s before and 2 s after the disturbance. The central 9–17 s portion is used only for the valve-torque decomposition in (14). Performance values are computed from the logged time series using the same time-weighted RMS definition,

$$x_{\mathrm{RMS}} = \sqrt{\frac{1}{t_b - t_a}\int_{t_a}^{t_b} x^2(t)\,dt}. \tag{24}$$

Support motion is measured by the deflection Δq_s from its value at valve-turn onset. Valve tracking is measured by e_θ, and motion outside the valve-progress direction is summarized by the projected error $\|\bar{\boldsymbol{e}}_{\perp}\|$ in mm-equivalent units. The wrist-force residual is defined as the norm of the difference between the measured wrist-force vector and a baseline estimated during the open-gripper stabilization period before closure. The baseline is kept fixed during the valve turn, and the reported peak is the maximum residual over the 5.5–21 s analysis interval.

V. RESULTS

A. Support motion and valve tracking

The prescribed support torque produces similar support trajectories across the controller configurations, as shown in

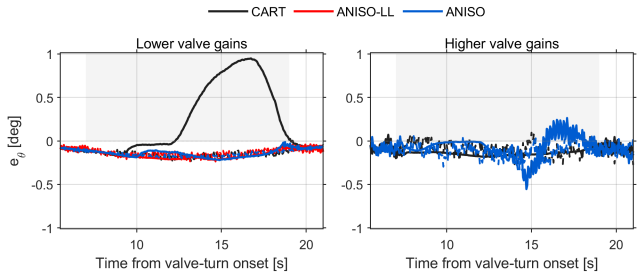


Fig. 3. Valve-tracking error for the lower and higher valve-direction gain settings. Dashed lines denote nominal cases and solid lines disturbed cases.

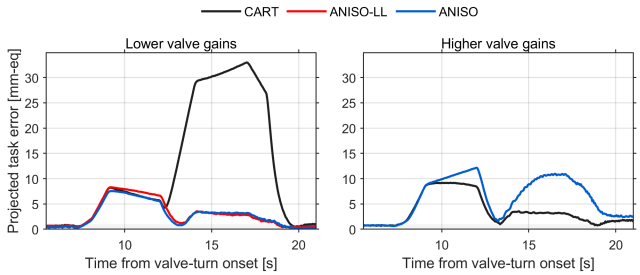


Fig. 4. Task-space error orthogonal to valve progress under the external support disturbance.

TABLE II
PERFORMANCE OVER THE 5.5–21 S ANALYSIS INTERVAL.

		Support	Valve tracking		Complementary error	
Controller	Condition	RMS (deg)	RMSE (deg)	Peak (deg)	RMS (mm-eq)	Peak (mm-eq)
CART-L	Nominal	0.298	0.147	0.232	0.680	0.977
CART-L	Disturbed	2.026	0.444	0.949	17.547	32.969
ANISO-LL	Nominal	0.296	0.145	0.238	0.734	1.076
ANISO-LL	Disturbed	2.113	0.159	0.219	4.067	8.325
ANISO-L	Nominal	0.297	0.145	0.206	0.434	0.626
ANISO-L	Disturbed	2.069	0.145	0.221	3.695	7.549
CART-H	Nominal	0.293	0.119	0.271	1.926	3.069
CART-H	Disturbed	2.063	0.135	0.199	4.732	9.197
ANISO-H	Nominal	0.293	0.136	0.341	2.055	3.255
ANISO-H	Disturbed	2.073	0.138	0.554	7.409	12.186

Fig. 2. Table II shows that the disturbed support-deflection RMS remains close to 2° in all five cases. The tracking comparison is therefore performed under similar disturbance-induced support motion.

With the lower valve-direction gains, CART-L, ANISO-LL, and ANISO-L have similar nominal valve-tracking error. Under disturbance, the tracking RMSE is $0.444°$ for CART-L, $0.159°$ for ANISO-LL, and $0.145°$ for ANISO-L. Thus, the reduction observed with the task-aligned formulation remains pronounced when the complementary gains are also kept at the lower level. Increasing the complementary gains from ANISO-LL to ANISO-L provides a smaller additional reduction. When the valve-direction gains are increased, CART-H and ANISO-H show similar disturbed RMSE values, $0.135°$ and $0.138°$, respectively. ANISO-H, however, reaches a larger peak tracking error, $0.554°$ compared with $0.199°$ for CART-H.

B. Response outside the valve-progress direction

Support motion also perturbs task directions that do not contribute to valve rotation. Figure 4 shows the norm of this projected task error during the disturbed condition.

With the lower valve-direction gains, projected task-error RMS is 17.547 mm-eq for CART-L, 4.067 mm-eq for ANISO-LL, and 3.695 mm-eq for ANISO-L. Most of the reduction therefore remains when the gains in the remaining task directions are also kept at the lower level. With the higher setting, CART-H and ANISO-H reach 4.732 mm-eq and 7.409 mm-eq, respectively.

For the Cartesian controller, error in the projected task subspace can also contribute to the valve-direction torque through (14). Over the 9–17 s central portion, the leakage term has RMS values of 1.27 mN m and 1.21 mN m for the lower- and higher-gain Cartesian controllers. These values correspond to 7.6% and 20.0% of the feedback torque generated by valve-tracking error alone during the same interval. The ANISO formulation removes this contribution through the projection in (17). Because the leakage is small in absolute magnitude, it does not by itself explain the differences in valve tracking. The ANISO-LL results further show that the lower tracking error persists when the gains in the remaining task directions are also kept at the lower level.

The coupling introduced by the Cartesian formulation is not limited to the leakage term in (15). Because the diagonal Cartesian gain matrices are not generally aligned with the valve-progress direction, the valve-tracking component can also generate feedback in the projected complementary subspace through terms of the form $\boldsymbol{P}_\perp \bar{\boldsymbol{K}}_j \bar{\boldsymbol{\xi}}_v e_\theta$ and $\boldsymbol{P}_\perp \bar{\boldsymbol{D}}_j \bar{\boldsymbol{\xi}}_v \dot{e}_\theta$. The task-aligned formulation separates these two feedback branches explicitly. The individual contribution of this additional coupling was not isolated in the present simulations.

In the disturbed lower-gain cases, the peak wrist-force residual was 1.814 N for CART-L, 1.264 N for ANISO-LL, and 2.389 N for ANISO-L. The valve-channel scale remained at $s_v = 1$ in all simulations, and the largest absolute arm-joint command was 17.142 N m. No arm-torque saturation occurred. Increasing the complementary gains from ANISO-LL to ANISO-L therefore produced a small additional reduction in tracking error, together with a higher peak wrist-force residual.

VI. DISCUSSION

The results show that the effect of unmodeled support motion depended on both the feedback structure and the distribution of impedance across the task directions. At the lower valve-direction setting, CART-L and ANISO-LL use the lower gain level also in the remaining task directions, whereas ANISO-L uses higher complementary gains. Most of the reduction observed with ANISO-L remains with ANISO-LL, indicating that the higher complementary gains account for only a smaller additional reduction.

CART-L and ANISO-LL have similar nominal tracking and use the same gain level in the valve-progress and remaining task directions, but differ in how these task errors enter the controller. CART applies one Cartesian impedance to the

combined task error, whereas ANISO regulates valve progress separately and projects the remaining feedback outside this direction. The small leakage term found for CART does not by itself explain the observed tracking difference.

At the higher valve-direction setting, CART-H and ANISO-H have similar disturbed tracking RMSE, while ANISO-H exhibits a larger tracking peak and projected task error. The effect of the task-aligned formulation therefore depends on the selected gain level and does not produce the same response across all settings. Because the absolute valve-tracking errors remain small relative to the 88° commanded rotation, these results should be interpreted as differences in closed-loop disturbance response rather than as demonstrated differences in task success or grasp stability.

The evaluation was conducted in CoppeliaSim with robot dynamics, contact interaction, passive support dynamics, and actuator limits included. These elements capture the robot environment dynamics considered in the present study, but the results remain limited to simulation. The study considered one support axis, disturbance profile, valve geometry, and characteristic length L_x, with one deterministic simulation per controller and support condition. The study is also limited to a diagonal Cartesian impedance baseline.

The controller does not update the valve pivot or axis measured at turn onset. The valve twist and projector use the current TCP position relative to this fixed geometry, which also defines the reference reconstruction. Changes in the actual pivot and axis caused by support motion therefore enter as unmodeled geometric perturbations, consistent with the assumption that the support state is unavailable to the arm controller.

Although the Cartesian gain matching is defined at the initial geometry, an offline evaluation based on the simulated moving valve geometry showed that the effective Cartesian valve-direction stiffness and damping remained within approximately 0.3% of their target values in the disturbed cases. The sensitivity to L_x was not evaluated; because this parameter affects the relative weighting of translational and rotational components, different values may change the quantitative response. Sensor noise and communication delay were also not varied. Future work should include sensitivity to L_x and gain ratios, broader disturbance conditions, repeated trials, and hardware experiments.

VII. Conclusion

This study evaluated anisotropic impedance for valve manipulation while an unmeasured passive support coordinate was driven by an external torque. The same disturbance produced similar support motion across the controller configurations, allowing the controllers to be compared under similar support motion. With lower gains along the valve-progress direction, CART-L, ANISO-LL, and ANISO-L showed similar nominal tracking. Under disturbance, their tracking RMSE values were 0.444°, 0.159°, and 0.145°, respectively. The lower error of the task-aligned controller therefore persisted when the gains in the remaining task directions were also kept at the lower level. Increasing those gains produced a smaller additional reduction in tracking error, together with a higher peak wrist-force residual.

When the valve-direction gains were doubled, the tracking difference became small and the two controllers exhibited different transient and projected task responses. ANISO-H also produced a larger tracking peak and projected task error than CART-H. The effect of the anisotropic formulation was therefore dependent on the selected gain distribution rather than indicating a general superiority over diagonal Cartesian impedance.

Further work will examine additional disturbance conditions, sensitivity to the task-space scaling and gain ratios, repeated trials, hardware implementation, and coordination with whole-body control.

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